

Engineering Community Resilience: A Transdisciplinary Early Career Researcher's Journey in Cultivating Bottom-Up Energy Initiatives Amidst Uncertainty

Vanja DJINLEV¹

Transdisciplinarity Lab (TdLab), ETH Zurich, Switzerland

Abstract. Navigating the evolving landscape of energy systems demands a multifaceted approach that combines engineering expertise with a holistic understanding of societal, economic, and environmental considerations. Early career researchers (ECRs) play a crucial role in bridging these disciplines, fostering innovation, and driving sustainable energy transitions. This paper delves into the unique intersection where an ECR in the engineering field transcends traditional technical boundaries, venturing into the realm of social sciences to foster energy communities. By employing a transdisciplinary approach, ECRs are not only contributing to the technical advancements in energy systems but are also actively engaging in community leadership and development, as tenets of energy communities. The role of engineers in this context is redefined – from mere providers of technological solutions to facilitators and catalysts of community-driven change. However, cultivating these initiatives is not without its challenges. The paper investigates the various forms of uncertainty that pose significant barriers to the development of bottom-up energy initiatives. These include regulatory ambiguities, market fluctuations, evolving technological landscapes, and financial constraints. Central to this paper is a case study from North Macedonia – the first energy community in the country, which provides concrete insights into the practical aspects of building and sustaining energy communities in a transitioning energy landscape. This case study not only illuminates the challenges faced but also showcases the strategies and approaches employed to overcome them. It serves as a microcosm of the broader global challenge in transitioning to more resilient, community-driven, and sustainable energy systems.

Keywords. Transdisciplinary engineering, early career researchers, bottom-up energy initiatives, community resilience, uncertainty

Introduction

At the heart of the energy transition is the need for a speedy uptake of vast renewable energy technologies (RETs) aimed at substituting the energy production from the dominant fossil fuel technologies and satisfy the ever-increasing energy demand [1]. Regarded as one of the ways for decreasing the devastating consequences of climate

¹ Corresponding Author, Email: vanja.djinlev@usys.ethz.ch

change, the uptake of RETs and their integration within existing energy infrastructures transcends technical boundaries and demands the integration of engineering, societal, economic and environmental considerations. Central to the integration of these aspects are researchers whose roles have evolved beyond conventional academic disciplines. For example, engineers that venture in the realm of social sciences might contribute to the speedy implementation of the much-needed renewable energy as they are better equipped to consider issues beyond the traditional engineering challenges. Along with the integration of multiple disciplines, engineers may engage in transdisciplinarity – a reflexive research approach that enables learnings between science and society [2]. By engaging in transdisciplinary (TD) research, engineers may go beyond the integration of disciplinary knowledge and engage with communities to solve a certain problem, while critically reflecting on their roles and positionality.

The integration of RETs entails a shift in the way the energy system is set up and operated. The accessibility of decentralized energy technologies with varied scales provides the opportunity for citizens not only to take direct action in the production and consumption of energy, but also to influence the aspects of ownership and decision making related with the energy technologies [3], aspects which were and continue to be quite entrenched. Decentralized energy technologies also provide citizens the opportunity to band together and form collectives centered around some energy aspect, including the production and consumption of energy, promotion of efficient energy practices, and bottom-up knowledge sharing. One such collective form is the energy community (EC), where participation is open and voluntary, having environmental and social objectives.

This paper aims to elaborate on the potential that transdisciplinary research may play in developing sustainable solutions for speeding up the energy transition. The paper may be regarded as a self-reflection of the author's own experience as a TD engineering early career researcher (ECR). At the heart of this study is a case from North Macedonia, a country where the author grew up, that will serve as a practical example and illustration of the challenges faced when setting-up an EC within a transitioning energy landscape. The paper will provide the practical steps taken to cultivate the first bottom-up, citizen-led energy initiative in the author's home country, aiming to improve the resilience of the community amidst uncertainty. Through a critical reflection of the steps, the paper will outline the variety of issues and learnings that TD engineering ECRs may expect to experience when initiating bottom-up energy initiatives.

The remainder of the paper is structured as follows. Chapter 1 outlines the role of TD in developing sustainable societal solutions, while Chapter 2 sheds light on how bottom-up energy initiatives may contribute to increasing communities' resilience. Chapter 3 provides the details of the context in North Macedonia, followed by the methodology shown in Chapter 4. The results are given in Chapter 5, while Chapter 6 discusses and reflects on the study.

1. Role of TD research in developing sustainable societal solutions

Transdisciplinary research has become a critical paradigm for tackling complex and multifaceted problems that cannot be sufficiently addressed by traditional disciplinary approaches, whether used in isolation or in combination. Sustainability challenges, such as climate change, social inequalities, and resource depletion, transcend disciplinary boundaries and demand collaborative, knowledge-integrating approaches [4].

Transdisciplinary research goes beyond the blending of multiple disciplines (multidisciplinarity) or the cooperation of several disciplines (interdisciplinarity), as it relates to ‘purpose-driven collaborative processes of knowledge production among researchers of different disciplines, inter- and transdisciplinary fields, and representatives of private and public sectors including civil society’ [5].

At its core, transdisciplinary research processes are reflexive [2], problem oriented and have a participatory, co-productive nature [6]. These co-productive processes empower various actors to share their expertise and insights, breaking down the traditional dichotomy between knowledge production and application. Transdisciplinary approaches recognize that problems are situated within real-world contexts and necessitates the integration of practical knowledge, interests and values as essential to solving the problems [7]. As such, the solutions derived from transdisciplinary interventions are often more robust, contextually sensitive, and have greater buy-in from the communities affected by the problem.

However, implementing transdisciplinary research presents unique challenges. Collaboration between researchers from vastly different fields, along with societal actors, requires fostering a culture of mutual respect, effective communication, and a willingness to navigate different epistemologies [8], and it requires significant effort in terms of coordination and communication between individuals [9]. The success of such initiatives often depends on the ability to manage these complexities effectively, highlighting the need for frameworks and tools that can facilitate transdisciplinary collaboration.

2. Bottom-up energy initiatives for increased community resilience

Decentralized energy technologies enable direct citizen participation in the energy transition, unlocking new forms of citizen engagement in the system, both as individuals and as groups. People are setting up or joining ECs, they are pooling resources to install RETs, coordinate their energy consumption and production [10], collectively invest in retrofitting [11] and increasing the energy efficiency of their homes and community buildings, as well as investing for the sole purposes of financial profits [12]. To this end, policy makers have implemented a wide range of strategies to empower individuals and communities in becoming more ‘active’. From disseminating information to providing access to financial incentives such as tax credits and subsidies, policy makers have been instrumental in supporting individuals and communities to uptake and democratize the access to new RETs. The European Commission for example has introduced directives that recognize Citizen Energy Communities (CEC) and Renewable Energy Communities (REC) as crucial supporting factors of the energy transition. Aside from increasing the uptake of RETs, both types of communities are expected to have positive environmental, economic and social impacts on all members [13].

As the number of ECs continues to increase, their contribution in reshaping the system and expanding the capacity of RETs is becoming more evident. The uptake of RETs enables the distributed production and consumption of electricity without relying on the vast electrical grid systems. In other words, electricity can be produced and consumed within the community, safeguarding the community from technical issues of congestion or perturbation of the electrical grid outside of the community. Therefore, ECs may better guarantee security of supply and well-being of community members in

remote areas that suffer from energy disruptions while at the same time improving the access to RETs.

When discussing the resilience of a community from an energy engineering perspective, the Energy Resilience (ER) framework developed by Arup serves as a basis [14]. This framework enables the determination of challenges and opportunities for improvement of energy systems, regardless of their size. Considering the applicability of the framework across energy systems of different sizes, in this paper, the ER framework is adapted to correspond with the features of a small rural community that seeks to improve its energy resilience. The adaptation entailed a selection of specific variables from the three dimensions of the ER framework, reframed as a community's ability to: i) minimize the duration and impact of energy disruptions, ii) maintain energy security and well-being during disruptions and energy price fluctuations, iii) improve access to renewable energy technologies and iv) improve energy efficiency. The variables of the framework were chosen based on the ECR's knowledge of the local context, along with the goals of improving sustainability and environmental protection activities. Despite choosing technical variables from the ER framework, the expected benefits go beyond the technical realm and include also positive social and environmental impacts.

Advancing the resilience of communities from an energy engineering perspective has significant climate benefits. When the duration and impact of energy disruptions are minimized, communities reduce their consumption of fossil-fuel-dependent generators and stoves, thus minimizing indoor and outdoor pollution. Maintaining energy security through diversification of energy sources further reduces the consumption of fossil-fuel-dependent technologies and it contributes to safeguarding the community from fluctuating energy prices. Moreover, improving access to renewable energy technologies directly reduces greenhouse gas emissions and further stabilizes price fluctuations, providing potential income source for the community from the sales of excess electricity to the grid. Lastly, improvements in energy efficiency reduce energy demand and expenditure, further reducing the greenhouse emissions and strengthening the income stream of the community.

3. Setting up the context: North Macedonia

Nestled in the heart of the Balkan peninsula, North Macedonia confronts a complex energy landscape, as it seeks to diversify its carbon-intensive energy sector. The country relies primarily on low-grade lignite coal for its electricity production (39.7%), followed by natural gas (28.1%) and hydropower (26.4%), while biofuels and waste (20.7%) take the third place after coal (62.9%) and hydropower (14.2%) in terms of the overall domestic energy production. North Macedonia's energy needs are satisfied mostly by imports, as the energy import dependence reached more than 60% in 2021 [15].

Faced with these challenges, the country actively pursues the diversification of its energy mix as the pillar of its energy transition goals. Phasing out coal and increasing the share of renewable energy are the country's main concerns, as the electricity production from renewables reached 26.5% in 2022. To phase out coal by 2030, the government launched projects for the construction of gas/hydrogen-fired power plants aimed at replacing the existing coal infrastructure [16].

As a candidate country for the European Union (EU), North Macedonia is currently harmonizing its national legislation across different sectors, including energy and environment. The harmonization of the legislation is expected to attract foreign

direct investments and modernize the energy sector, while the integration in EU's internal energy market is expected to increase the energy security of the country [17]. In terms of direct citizen participation in the energy system, the country has enabled households to install up to 6 kW capacity of rooftop solar power plants and to sell the surplus of electricity. However, there is no national legislation concerning energy communities or any types of bottom-up citizen-led energy initiatives. This lack of legislation is the main barrier to the creation of energy communities or any other type of citizen-led energy initiative. The following chapter provides the methodology behind creating the first energy community in North Macedonia amidst the uncertainty of having no legislation.

4. Research strategy

The creation of the first energy community in North Macedonia was a long process that included different intertwined tasks that went beyond the engineering domain. As a TD activity, the creation of the energy community was grounded in the cross-section of energy engineering and social sciences, integrating the knowledge from both academic and non-academic sources. The methodology included a mixed methods data collection approach coupled with the necessary techno-economic energy modelling study. Despite the tasks being intertwined, there were four distinctive steps that led to the creation of the energy community, without considering the installation of the equipment as the final necessary aspect.

Step 1: Field research

The first step included a visit to the community to understand the situation on the ground. The community is located in Belitsa, a rural area in the region of Kichevo, with a population of around 100 people. When visiting the area, particular attention was paid to the types and conditions of the buildings present in the community, the layout of the area and its surroundings, the energy sources used for different purposes, the energy infrastructure in place, and the potential natural resources to be used as energy sources.

Step 2: Workshop - The ENCLUDE Academy for Energy Citizen Leadership

After determining the current situation on the ground, the next step was to understand the problems and needs of the community in terms of energy, as well as to co-create the potential solutions together with the members of the community. For that purpose, the program that was developed and used in the context of the ENCLUDE Academy for Energy Citizen Leadership as part of the H2020 ENCLUDE project, was also used to set up the energy community in Belitsa. Created as an online leadership development and civic engagement program, the ENCLUDE Academy aimed at helping citizens interested in making changes in their own communities to take local, bottom-up action in support of decarbonization. The program was structured around design thinking for systemic transformation [18], and it was delivered in more than 40 hours over the course of six months.

In the context of Belitsa, the ENCLUDE Academy program was adapted and delivered as a two-day workshop, retaining the original structure. This means that the community members were able to go through the first three steps of the design thinking methodology (empathizing, defining and ideating) of the ENCLUDE Academy program through an intense work process. The workshop kicked-off with the creation of a rich

picture as an initial step for creating common understanding and bringing all residents on the same page while enabling individual perspectives of the context to be equally represented. The last two steps of the design thinking methodology (prototyping and testing) were done together with the facilitator of the workshop, i.e. the author of this study. Considering the lack of technical/engineering competences of the participants, the facilitator provided the contextual boundaries (including but not limited to the technical features of RETs and relevant regulatory frameworks) within which technical solutions were proposed. Taking this approach, the facilitator ensured that the participants could go through all phases of the design thinking methodology.

The program was attended by members of the rural community Belitsa and members from three neighboring communities, along with a representative from the Kichevo municipality and representatives from non-governmental organizations working in the local community.

Step 3: Interviews with community members

The next step of the methodology was pivotal in gaining in-depth insights into the community's expectations regarding the technical solutions determined in the previous step. By engaging directly with community members, this step aimed to delve beyond the understanding from the previous steps, uncovering the nuances of how the community interacts with energy on a daily basis. These interactions served not only to identify the specific technological solutions that would best meet the community's needs but also to understand the broader expectations from such interventions. This approach ensured that the proposed energy technologies would be well-aligned with the community's expectations and energy practices, fostering a sense of ownership and acceptance among the members.

Understanding the community's energy practices and norms through these interviews was crucial for tailoring the energy solutions to fit seamlessly into the existing context. Furthermore, the discussions around financing shed light on the economic landscape of the community, revealing the financial capabilities and related constraints faced by the members. This knowledge was vital for designing financially viable and accessible energy solutions that fully align with the socio-economic fabric of the community.

For this reason, interviews were organized with the three official community representatives (the community leader, the person responsible for finances and administrative issues, and the person responsible for outreach and public relations). The interviews lasted between 50 and 70 minutes and they were audio recorded. The recordings were used for creating interview transcripts that were stored, in an anonymized form, on the researcher's computer. Considering the low number of interviews, no interview analysis software was used and instead, the coding was done by the researcher.

Step 4: Techno-economic analysis

After determining the energy demand and learning about the energy practices, expectations of the community in terms of energy technologies and the financial situation, the final step included the techno-economic analysis of potential energy solutions. More specifically, the feasibility study aimed at determining the technical solutions with the lowest Levelized Cost of Electricity (LCOE) and the highest possible penetration of RETs. For that reason, the HOMER software was used, which is a tool used for designing and analyzing microgrids and distributed energy projects. HOMER allows the

optimization of the mix of renewable and conventional sources in a power system to meet specific objectives, such as minimizing cost, reducing emissions, or improving reliability. HOMER simulates the operation of a system by evaluating a wide range of technology options and identifying the most cost-effective and efficient solution for the specified operating conditions. It assesses the interaction of energy technologies and models their economic and technical feasibility by considering factors like variability in renewable resource availability, energy demand patterns, and costs associated with installing and operating different technologies.

5. Overview of the findings

The field research (step 1 of the methodology) aimed at determining the layout of the area, the types and conditions of the buildings, the energy sources that are currently used, along with the existing electrical infrastructure and the potential for developing and using renewable energy sources.

Through the adapted ENCLUDE Academy program, step 2 of the methodology aimed at determining the problems and needs of the community in terms of energy, as well as to co-create the potential technical solutions together with the members of the community. In this step, the TD engineering ECR played a crucial role as a facilitator in the workshop, guiding the participants through the existing energy regulatory landscape towards feasible technical solutions that may be applicable in the given context, based on the learnings from the field research. By following the ENCLUDE Academy program, the following problem statements were coined:

- PS1: The rural area [of Belitsa] needs a stable supply of electricity with no interruptions (blackouts or brownouts), however the electrical grid of the village is old, has low capacity, and has limited interconnectedness with the villages in the vicinity.
- PS2: Residents [of Belitsa] want to increase their thermal comfort and decrease their energy costs by changing their current biomass heating systems with more efficient systems, however they cannot rely on air-to-air heat pumps due to the old electrical grid in the village.
- PS3: Residents [of Belitsa] need to modernize their current social gathering center to be usable throughout the year, however due to the current biomass heating system and the high energy expenditures associated with it, the center is not used in winter.

After defining the problem statements, the next step revolved around the ideation of potential solutions in respect to the identified problems. Prior to this step, the facilitator gave a brief overview of technologies which might be applicable to the context, focusing on RETs. The following technical solutions were proposed:

- Change the area's electrical grid into a fully off-grid system powered by RETs only,
- Use the river to install a second hydropower plant,
- Install solar PV on the social center,
- Install solar PV on the place of worship,
- Install solar PV on all houses in the area,
- Install air-to-air heat pumps in all buildings,
- Use a single, community-owned geothermal energy technology,
- Construct community-owned wind turbines,

- Use a biomass-based combined heat and power (CHP) plant for the area,
- Develop the agriculture in the area with agri-photovoltaics (agri-PV).

The last step of the methodology included the techno-economic analysis of a system that corresponded to all input parameters collected in the previous steps. The available finances of the community were the main constraint that influenced the choice of the idea (technical solution) elaborated in the previous steps. After considering the collected data, the technical solution included a combination of a few of the proposed ideas, and the project implementation was divided into five phases, with different start dates and activities. The first phase involved the installation of a solar PV system on the social center with a combination of air-to-air heat pumps. The combined system includes a 20-kW solar PV with two 7.5 kW air-to-air heat pumps, and a bidirectional 40 kW smart meter. The second phase is the expansion of the solar PV system, with an additional 20-kW solar PV system. The total installed capacity of 40 kW solar PV on the social center will not be expanded in the future, due to the legal constraints regarding energy self-production on a single (connection) point by legal entities. Therefore, the subsequent phases include the expansion of solar PV systems across the community but at different connection points to the grid, keeping up with the 40-kW capacity constraint. These systems will be installed on individual households throughout the community, as well as on communal agricultural spaces, with the idea of creating the first agri-PV instance in the country.

6. Discussion and reflection

The creation of the first energy community in North Macedonia, as presented in this paper, serves as evidence of the role that TD engineers can play in driving sustainability transitions through bottom-up initiatives. This case study demonstrates how the proposed technical solution supports the community's resilience from an energy engineering perspective, as it contributes to the minimization of the energy disruptions due to the locally produced electricity. The locality of the production also provides benefits for the energy security and well-being of the community, as during disruptions the solution maintains the access to electricity and safeguards against grid price fluctuations. The installation of solar PV systems improves the access of the community to RETs, while the proposed activities contribute to the overall energy efficiency improvement. In general, the proposed technical solution has positive decarbonization effects, aligning with the country's broader environmental objectives and climate commitments.

The TD approach adopted throughout the process was instrumental in developing the bottom-up solution. Aside from considering engineering solutions, the approach considered the social, economic, environmental, and behavioral contexts of the community in relation to energy. In essence, the overall approach recognized that local problems might not be confined to a single academic discipline, and that any sustainable solution provided needs to go beyond disciplinary boundaries and consider a collaborative approach. By integrating the academic and non-academic knowledge through a participatory process, the TD engineer helped in the co-creation of a bottom-up solution that was not only technically viable, but also economically feasible, environmentally sound, and socially acceptable. The integration of academic and non-academic knowledge is at the core of a TD project [5] aimed at creating sustainable

solutions to complex problems, and by taking this approach, the TD engineer ensured further sustainable community development.

Another important role that the TD engineer took throughout the co-creation process was to provide the boundaries within which ideas could be proposed. The boundaries relate to the specific uses of the technologies, their techno-economic aspect, their applicability in the considered context as well as the current policy landscape that governs them. As a facilitator of the co-creation process, the TD engineer ensured that the process did not get sidetracked or disrupted by ideas that went beyond the boundaries. The introduction and overview of the energy technologies that could be adopted in the specific context at the beginning of the co-creation process ensured that the community members had at least the same minimal understandings of how the technologies work and where they may be applicable.

Reflecting on the overall process, several important points must be outlined. First and foremost, knowledge of the local context is paramount. This relates not only to the regulatory landscape, but also to the social relations, needs and expectations of the community, as well as their understanding of energy-related issues, from practices to perceptions, as they influence the whole energy system. Furthermore, trust is essential for solutions to be accepted – in situations in which the TD researcher is an outsider of the community, building certain relationships with members of the community enables co-creation processes to run more smoothly. The building of trust and understanding between the community and the TD researcher also minimizes the knowledge imbalances that may morph into power imbalances, considering that the community has the final say across the whole process. Since the role of the TD engineer goes beyond the standardized, engineering roles of providing technical recommendations and instead entails the interaction with the community for joint problem-framing and knowledge co-creation, it is essential for the TD engineer to bring energy topics closer to the community in an understandable manner and ensure a leveled field for interaction.

One important asset that the TD engineer can utilize throughout this process is the local leader, or the person who pushes the ideas forward within and represents the community. Since the overall TD process is resource intensive (in terms of time, financial and human resources), having the local leader informed and supportive of the overall process helps in establishing the needed relationships and building trust with the community while minimizing any occurrences that may slowdown the overall process. The local leader will also be able to take care of any organizational and administrative aspects as well as mobilize human resources needed for setting up and future growth of the community beyond the aforementioned steps and ensure the overall running of the energy community.

Lastly, the uncertainties related to the lack of legislation and the evolving energy landscape are challenges that may hinder the implementation and/or the lifespan of the project. This is even more challenging when it comes to TD ECRs, as aside from the obligations for academic advancement, building strong foundations in multiple disciplines and effectively bridging disciplinary divides, having to implement and foster bottom-up initiatives – an aspect that vastly depends on regulatory landscapes, further strains TD ECRs resources and capacities. Therefore, it's advisable that these researchers engage with cultures and contexts in which they previous ties or have certain familiarity with, in order to minimize the resources needed to develop context-relevant solutions.

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